CALIBRATION CERTIFICATE

CERTIFICATE NUMBER	017851185163
PYRHELIOMETER MODEL	SHP1-A
SERIAL NUMBER	185163
SENSITIVITY	8.52 μV/W/m²
TEMPERATURE	22 ± 2 °C
REFERENCE PYRHELIOMETER	Kipp & Zonen CHP 1 sn REF1 active from 01 August 2018
CALIBRATION DATE	28 September 2018

Calibration procedure

Exact interchange of test pyrheliometer and reference pyrheliometer in a horizontal parallel beam of light from a Xenonlamp. Full collimation angle of beam is 1.0°. Minimum irradiance 650 W/m². Room temperature 22 ± 2 °C.

Hierarchy of traceability

This reference pyrheliometer was compared against the absolute cavity pyrheliometer PMO6 SN 103 using the sun as source according to ISO 9059 "Calibration of field pyrheliometers by comparison to a reference pyrheliometer". The reference PMO6 is calibrated against the World Standard Group (WSG), maintained at the WRC Davos every International Pyrheliometer Comparison (IPC). During the yearly IPC hosted by NREL in Boulder Colorado the reference PMO6 is verified. The readings are referred to the World Radiometric Reference (WRR) as stated in the WMO Technical Regulations. The originally estimated uncertainty of the WRR relative to SI is ±0.3%. The measurements were performed in Davos and Boulder Colorado (Davos: latitude: 46.8143°, longitude: -9.8458°, altitude: 1558 m above sea level; Golden: latitude: 39.742°, longitude: 105.18°, altitude: 1829 m above sea level). WRR- factor of PMO6: 0.99789 (from the last international Pyrheliometer Comparison, IPC-2015).

The comparison was performed in Tabernas, Spain (latitude:37.094°, longitude: -2.3547°, altitude: 503m above sea level). During the comparisons the reference pyrheliometer received direct solar radiation with intensities ranging from 767 to 938 W/m², with a mean of 890 W/m². The ambient air temperature ranged from 22.7 to 29.8 °C with a mean of 27.3 °C. The sensitivity calculation is based on 101 individual measurements. The sensitivity and its expanded uncertainty (95% level of confidence) are only valid for similar environmental conditions and amount:

7.95 ± 0.02 μV/W/m².

Date of measurements:2018, 8, 10-12 June

Justification of total instrument calibration uncertainty

The combined uncertainty of the result of the calibration is the positive "root sum square" of two uncertainties.
1. The expanded uncertainty due to random effects and instrumental errors during the calibration of the reference CHP 1 is
0.02 / 7.95= ± 0.25%. (See traceability text).
2. Based on experience the expanded uncertainty of the transfer procedure (calibration by non-simultaneous comparison) is estimated to be ± 1 %.
The estimated combined expanded uncertainty is the positive "root sum square" of these two uncertainties:
$\sqrt{(0.25^2 + 1^2)} = \pm 1.03$ %.

Notice

The calibration certificate supplied with the instrument is valid at the date of first use. Even though the calibration certificate is dated relative to manufacture, or recalibration, the instrument does not undergo any sensitivity changes when kept in the original packing. From the moment the instrument is taken from its packaging and exposed to irradiance the sensitivity may deviate with time. See the 'non-stability' value (% change in sensitivity per year) given in the radiometer specifications.

Delft, The Netherlands, 28 September 2018

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